

Beyond Latency: Understanding ONNX Execution on AI PCs

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ONNX Community Meetup 2026





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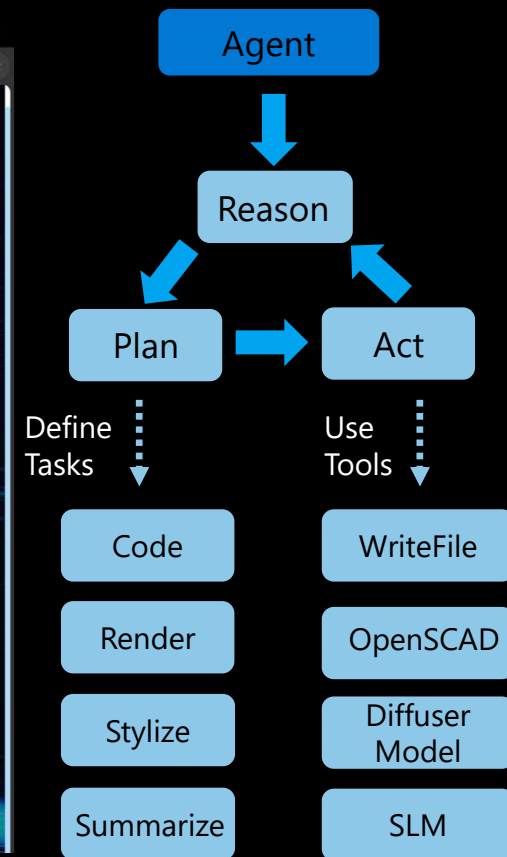
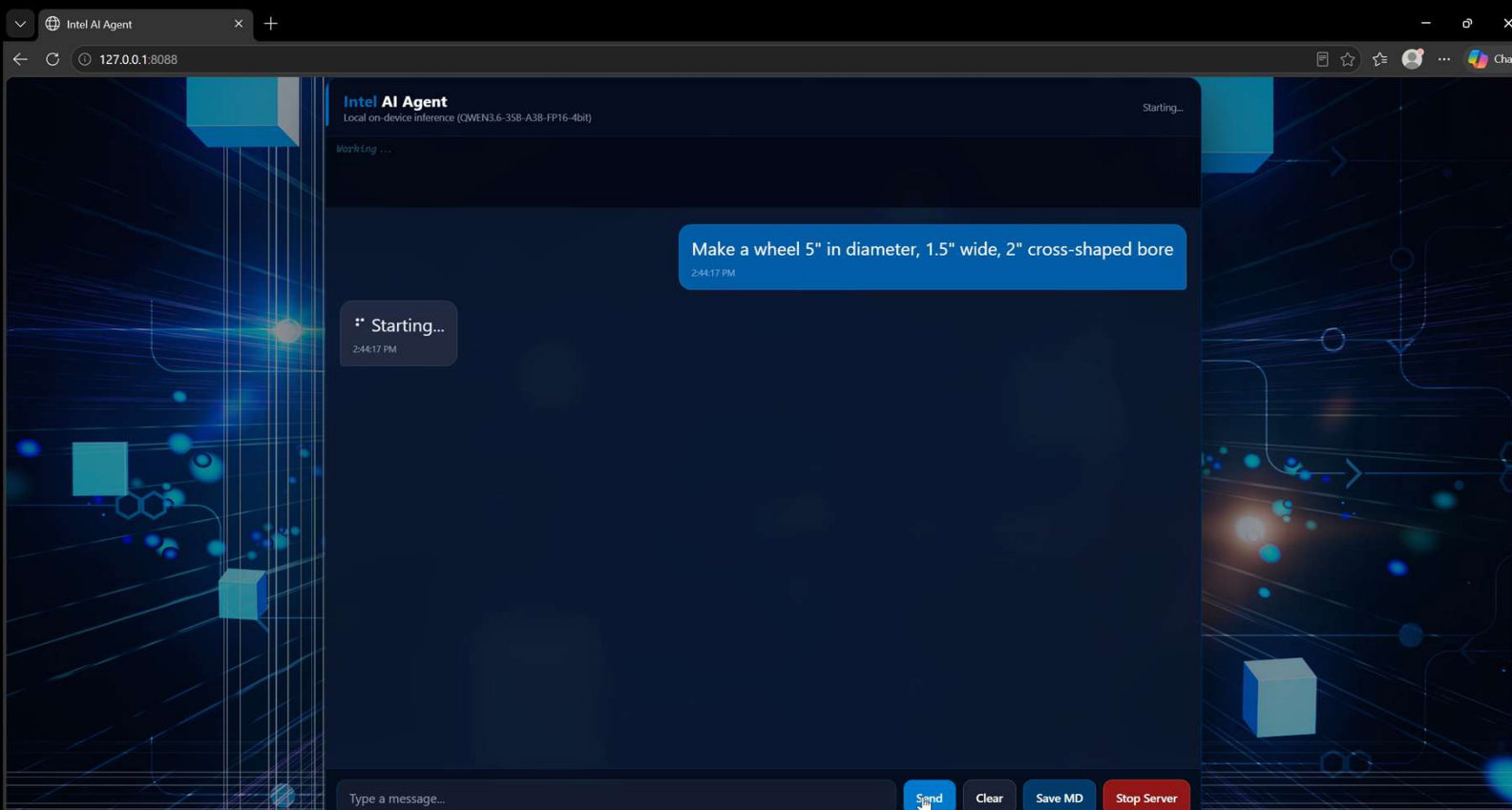
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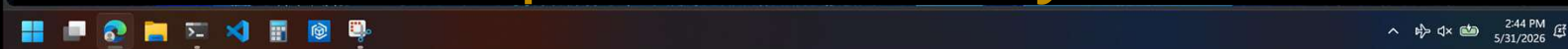
Agenda

- Understanding Agentic Profiling
- Intel® Unified Telemetry
- Intel® Instrumentation and Tracing Technology
- Profiling ONNX with ORT + OpenVINO EP
- Wrap up & Questions

Example Agent Workflow – 3D Printing



You cannot optimize what you cannot see



Understanding Software and System Behavior

2 categories of data to understand system behavior

Platform and Infrastructure

- Platform Telemetry
- CPU, GPU, NPU, Networking, Power
- Structured metrics system counters
- Usually low volume data at very low overheads

Application & Middleware

- Application logic visibility
- Unstructured logs, distributed traces
- High volume data at higher overheads

Understanding Software and System Behavior

2 categories of data to understand system behavior

Platform and Infrastructure

- Driver based
- Intel® Unified Telemetry
- CPU, GPU, NPU, Uncore, Power
- System through OS-specific tools

Application & Middleware

- Tracing through Intel® Instrumentation and Tracing Technology (ITT)
- ITT-enabled OpenVINO
- More layers are being enabled

Understanding Software and System Behavior

Correlation is the Key

Platform and Infrastructure

- Driver based
- Intel® Unified Telemetry
- CPU, GPU, NPU, Uncore, Power
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Need to bridge physical hardware performance with high-level software layers

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Intel® Unified Telemetry Approach (in **Preview***)

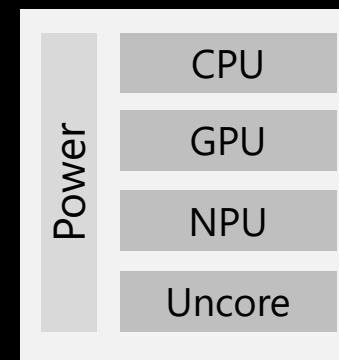
Status Quo

- Different hardware and software collectors tightly coupled with the tool
- Partial access to platform data (CPU, Power)
- Timestamp correlation can be a challenge, usually handled by the tools

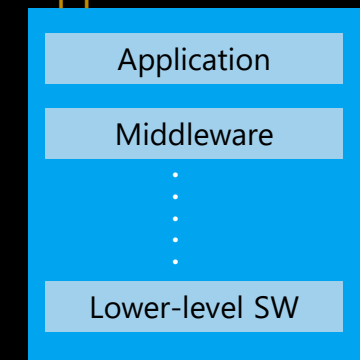


What we are moving to

Platform



Application SW



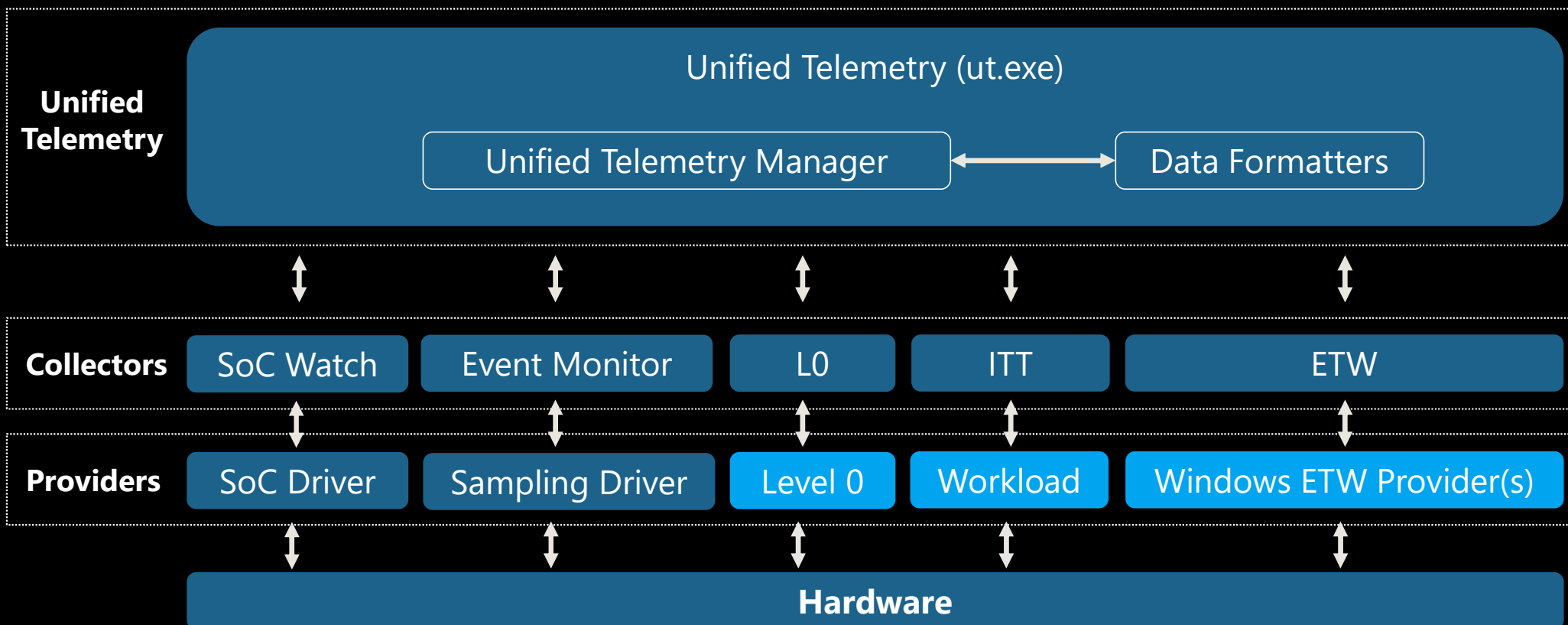
- Timestamp correlation for all streams
- Common format output
- Access to all collected data (platform + software)

Download <https://www.intel.com/content/www/us/en/developer/tools/intel-unified-telemetry.html>

Unified Telemetry Architecture

Intel

Intel or 3rd party



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Intel® Instrumentation and Tracing Technology (ITT)

Software Introspection

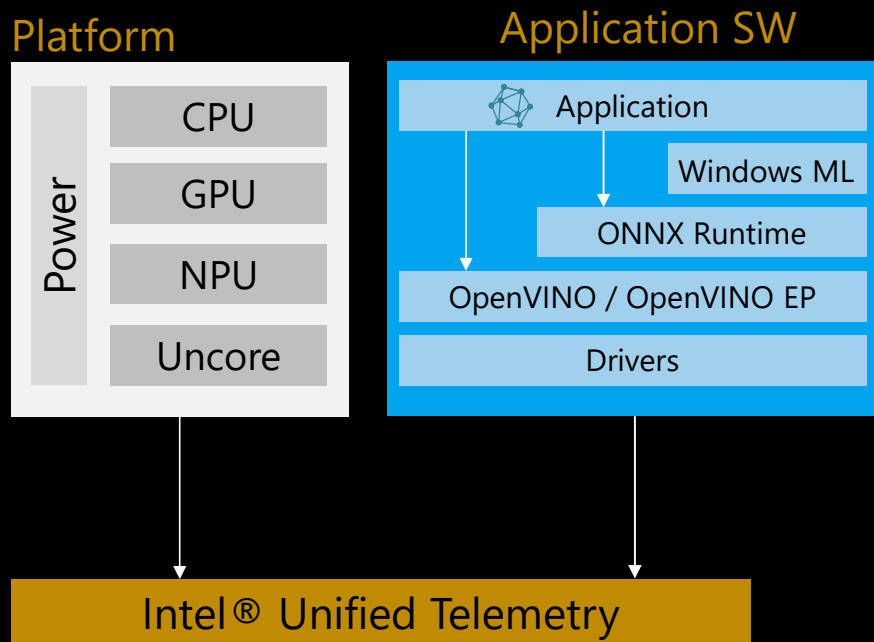
- Can be automatic through stack-walking
- Selective instrumentation
 - Needs to be low overhead and efficient
 - Provides knobs to control data flow
 - Generate human readable logical tasks

ITT Excerpts – Key API*

- Developer identifies code to instrument
- C/C++ API, Python bindings
- Domains – partition and organize
`__itt_domain_create()`
- String handles – eliminate overheads
`__itt_string_handle_create()`
- Execution Spans – algorithmic blocks
`__itt_task_begin()`
`__itt_task_end()`

*<https://github.com/intel/ittapi>

Intel® Unified Telemetry for your Tools



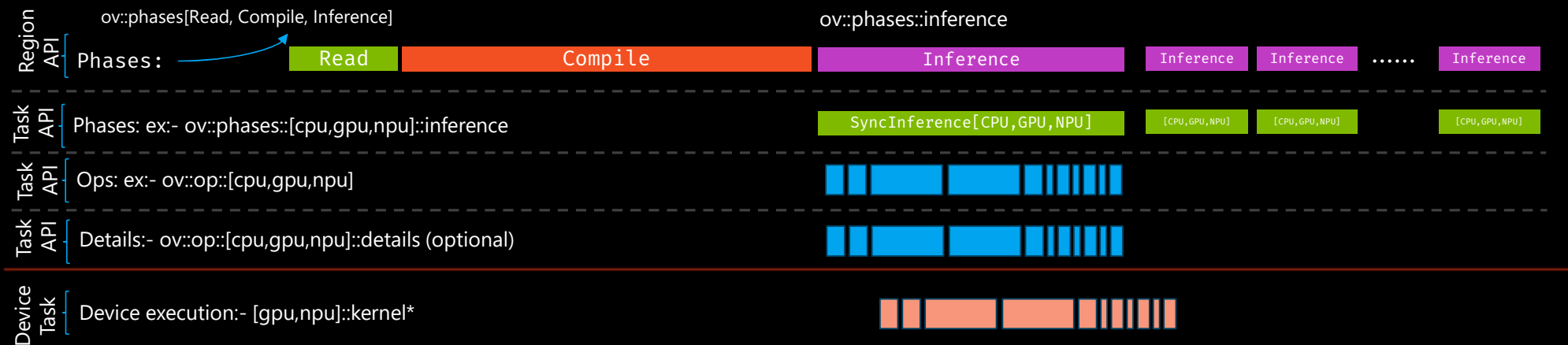
- OpenVINO is ITT-ready*, providing rich foundation for performance analysis
- OpenVINO EP** unlocks platform instrumentation with ORT & Windows ML APIs
- To correlate platform telemetry, just add traces to your app tools!

*Starting OpenVINO 2026.1

**Starting with latest EP released in May'26

ITT Schema for OpenVINO

- Top level time scopes use ITT region API (global)
- Nested level scopes use ITT task API (per-thread)
- Device side tracing is driver specific and normalized as a task (per device/per sub-device)



*Domain names not final, reflects GPU representation; feedback welcome

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BERT Test

bert-test.py

```
import itttapi
:
:
itt_domain = "bert-test.py"
task_tokenizer = itttapi.task("Tokenizer Load", itt_domain)
task_tokenizer.begin()

...
tokenizer = AutoTokenizer.from_pretrained(model_id)
...
task_tokenizer.end()

:
:
:
session = ort.InferenceSession(str(onnx_path), sess_options=sess_options)

:
:
:
```

Summary

Intel® Unified Telemetry takes a platform approach to profiling, providing consolidated hardware & software view

ONNX execution with ORT & WinML get enlightened view through instrumented OpenVINO inference stack

Call to Action:

Define standard to correlate executed model to source model

More Information

Unified Telemetry

- bit.ly/intel-unified-telemetry
- bit.ly/intel-itt

OpenVINO Plugin EP

- [pypi.org/project/onnxruntime-**ep**-openvino](https://pypi.org/project/onnxruntime-ep-openvino)
- [nuget.org/packages/Intel.ML.OnnxRuntime.**ep**.OpenVino](https://nuget.org/packages/Intel.ML.OnnxRuntime.ep.OpenVino)

OpenVINO

- docs.openvino.ai